

Antoniu Fodor, Ph.D.

antoniufodor.official@gmail.com | www.linkedin.com/in/antoniufodor

SUMMARY

Research data analyst with quantitative PhD and 8 years of experience designing and implementing end-to-end data analysis using Python, SQL, and Excel for projects on large, complex datasets. Skilled in building data pipelines, resolving data quality issues, and extracting analytical trends to support research projects. Experienced in communicating findings to stakeholders through recurring reports and presentations.

TECHNICAL SKILLS

Data Analysis & Programming: SQL, Python (NumPy, SciPy, pandas, Matplotlib, Seaborn, Plotly), MATLAB, IDL

Analytical Methods: Statistical Modeling, Regression, Probability, Exploratory Data Analysis, Diagnostic Analysis, Correlation Analysis, Hypothesis Testing, Spectral Fitting, Uncertainty Quantification, Data Wrangling

Data Visualization & Reporting: Power BI (DAX expressions), Microsoft Excel (Pivot Tables)

Developer Tools: Jupyter Labs/Notebooks, Git, Bash, Visual Studio Code, AI-assisted development (LLMs)

EXPERIENCE

Graduate Researcher

Aug. 2018 – Mar. 2026

University of Toledo

Toledo, OH

- Built and optimized automated Python data pipelines processing 600+ files across 100+ systems, improving data accuracy by reducing manual file handling, cutting runtime, and mitigating error propagation issues.
- Queried large-scale relational databases (200M+ records) using SQL to cross-validate pipeline outputs against archival datasets, identifying 8 anomalies and implementing corrective offsets to ensure data integrity.
- Conducted statistical analysis to identify primary drivers of system behavior across 100+ systems using Pearson correlation ($r = 0.85$), leading to a first-author publication.
- Identified systematic data quality issues in observational instrumentation through root cause analysis, directly informing the development of improved data cleaning process workflows by telescope staff.
- Developed Python and Excel visualizations and analytical reports presented at weekly group meetings, to cross-institution collaborators, and at 9 conferences, shaping observing strategy and research project direction.
- Managed 8 undergraduate researchers through weekly check-ins, deliverable tracking, and stakeholder reporting to senior leadership, resulting in 5 completed capstone projects, 1 completed NSF REU experience, and 1 forthcoming publication.

Graduate Teaching Assistant

Aug. 2018 – Aug. 2020

University of Toledo

Toledo, OH

- Instructed 100+ undergraduates across STEM laboratory courses and contributed to curriculum review modernizing lab materials and improving instructional clarity.

DOMAIN EXPERTISE

Spectrograph Technical Expert & Instructor

Apr. 2020 - Jun. 2025

- Served as primary technical resource for Lowell Discovery Telescope operations at the University of Toledo, supporting faculty, astronomers, and students during observing sessions and authoring tutorials for telescope usage and data retrieval.
- Maintained a shared data repository for collaborative projects including documentation and tutorials, supporting 11 researchers across 5 undergraduate projects and 3 graduate research projects.
- Designed and taught data cleaning class sessions in the Astronomy 4880 course to transform raw telescope data into informative data cubes for scientific analysis, adopted as the basis for a multi-week class project.

EDUCATION

University of Toledo

Toledo, Ohio

Ph.D in Physics

Aug. 2025

Albion College

Albion, MI

Bachelor of Arts in Physics and Mathematics (magna cum laude)

May 2018